

VALENTIN BERNADET

AI & Software Engineering Student | 6-Month End-of-Studies Internship from February 2027 | R&D · AI · Consulting
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SUMMARY

Final-year engineering student (Master's level) at ESILV who built and deployed a semantic search / RAG path inside an LLM-driven analytics service during an AI/ML engineering internship at IFS (Spain): Azure OpenAI embeddings, PostgreSQL/pgvector, deployed on Azure Kubernetes with Helm and CI. Personal projects in LLM/RAG, deep learning, WebGPU graphics and FPGA hardware; client-facing experience; trilingual French / Spanish / English. Seeking a 6-month internship from February 2027 in AI/R&D, software engineering or technical consulting, in France or abroad.

EDUCATION

École Supérieure d'Ingénieurs Léonard de Vinci (ESILV) – Paris-La Défense, France Sep 2022 – Sep 2027 (expected)

Master of Engineering (Diplôme d'Ingénieur) – Major: Creative Technology, Institute for Future Technologies

Relevant coursework: Machine Learning, Deep Learning, Human-Computer Interaction, Embedded Systems

Lycée Lakanal – Sceaux, France | French Baccalaureate (Math, Physics-Chemistry, Engineering Sciences), **Highest Honors** 2022

PROFESSIONAL EXPERIENCE

IFS – AI/ML Engineering Team (IFS Ibérica), Madrid, Spain

Apr 2026 – Aug 2026

AI / Machine Learning Engineering Intern

- Designed and built a **semantic search / RAG** path in GroundControl, an LLM-driven reliability analytics service (proof of concept): Azure OpenAI embeddings stored in **PostgreSQL/pgvector**, cosine-distance retrieval over free-text maintenance notes, answers grounded only in retrieved evidence.
- Routed 3 new intents through the existing LLM intent classifier (zero extra model call), with a non-regression test suite; extended the React/TypeScript front end to display retrieved notes and the answering path.
- Deployed on **Azure Kubernetes (AKS)** with **Helm and Docker** behind a feature flag; wrote 198 offline tests running in the Bitbucket CI/CD pipeline (SonarQube, Trivy); documented the feature in 3 Confluence pages.
- Worked in a distributed international team (English as working language) with Git topic branches, pull-request code reviews, Jira and Confluence.

PURENAT – Air-treatment startup, Biarritz, France

Jun 2024 – Sep 2024

Engineering Intern – Airflow Systems & Cybersecurity

- Analyzed security risks and implemented security protocols; contributed to airflow (aeraulic) engineering.
- Worked in a small cross-functional startup team with exposure to fundraising and patent filing.

TECHNICAL PROJECTS

- Research Paper LLM Agent** (LLM, RAG) – Web app comparing scientific papers with an LLM and RAG pipeline.
- 3D Scene Editor** (WebGPU, raymarching) – Interactive 3D scene editor with object configuration UI.
- MNIST Digit Recognition** (Python, deep learning) – Handwritten digit classifier deployed on the web.
- E-textile Capacitive Matrix (ongoing)** (FPGA, sensors) – E-textile using capacitive sensing matrix for tangible interfaces (HCI).
- The Extra Pocket (Kickstarter)** (product design, prototyping) – Upcycled-leather belt pocket, from concept to crowdfunding launch.
- Robotics & Mechanics** (CAD, electronics) – Medical octopod robot with custom electronics; mechanical ornithopter.

LEADERSHIP & ADDITIONAL EXPERIENCE

Vice-President, Student Union (BDE Chronos) – Pôle Léonard de Vinci, 3 engineering, business and digital schools 2024 – 2025

- Co-led event management across 3 schools, including major events for up to **3,000 attendees** with high-profile guests.
- Planned logistics and budgets for student trips and themed days.

Private Tutor (self-employed) – tutored students in math, planning lessons and tracking progress

2022 – 2024

Client-facing roles – Head Waiter, Surfing (Biarritz, 2025); Server, Michelin-starred restaurant, Hôtel du Palais (Biarritz, 2023); Lifeguard, Seagreen (Bidart, 2023)

- Served an international, high-end clientele and handled dozens of tables daily under pressure; made fast safety decisions responsible for hundreds of beachgoers.

SKILLS

AI & Data: Python, Large Language Models (LLMs), Azure OpenAI, RAG, embeddings, vector search (pgvector), semantic layers (Cube.dev), machine learning, neural networks, SQL/PostgreSQL

Cloud, MLOps & Methods: Microsoft Azure (AKS), Docker, Kubernetes, Helm, KServe, CI/CD (Bitbucket Pipelines), pytest, Git, Jira, Confluence

Software & Hardware: C#, TypeScript/React, FastAPI, HTML/CSS, WebGPU, Arduino, embedded systems, sensors, PCB (KiCad), SolidWorks, Fusion 360, Onshape, 3D printing

Languages: French (native), Spanish (native), English (professional working proficiency, B2+)

Interests: Rugby, tennis, triathlon, guitar, biotech, museums